

Henry Demarest

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EDUCATION

Stanford University

Master's Degree in Aerospace Engineering

- **GPA:** 3.96

Palo Alto, CA

Sept. 2025 – June, 2027

Yale University

BS in Mechanical Engineering (ABET), BS in Computer Science

- **GPA:** 3.98
- **Awards:** Summa Cum Laude, Phi Beta Kappa, Tau Beta Pi, L.C. Lichty & E.O. Watters Prize in Mechanical Engineering, Matthew Isakowitz Fellowship, Robert E. Lewis Award, Wesley M. Oler Prize
- **Non-Technical Activities:** First-Year Counselor, Vice Chair of Yale ASME, MechE Peer Mentor, Engineering Tour Guide, Yale Club Running, DPods Orchestra, Backpacking Trip Leader, Intramural Secretary

New Haven, CT

Aug. 2021 – May 2025

WORK EXPERIENCE

Relativity Space

GNC Engineering Intern

- Fully owned the development, tuning, and analysis of a novel flight algorithm to be used on TerranR's ascent
- Maintained and improved a monte carlo simulation case related to pad drift on launch

Long Beach, CA

June 2026 – Sept 2026

GNC Simulation Engineering Intern

- Developed an improved architecture for modeling rocket engine dynamics during flight
- Integrated an improved aerodynamics model for stage interactions during separation into high fidelity simulation

May 2025 – Aug 2025

GNC Performance Engineering Intern

- Led key trade studies focused on optimizing vehicle trajectory via different loads-informed throttle controllers
- Implemented automated feature to improve link between monte carlo data and trajectory optimization framework

May 2024 – Aug 2024

Astrobotic Technologies

Lunar Surface Systems - Mechanical Engineering Intern

- Designed test stands to subject components (solar arrays and dust covers) to cryogenic/vacuum environments
- Iterated upon the design of a spacerless-MLI thermal enclosure to enable spacecraft survival of the lunar night

Pittsburgh, PA

May 2023 – Aug 2023

PROJECTS

Yale Undergraduate Aerospace Association

Executive Board - President

- Led the executive board, and engaged with sponsors, administrators, aerospace companies, and other stakeholders
- Hosted inter-college conferences, industry recruitment events, volunteering opportunities, and technical workshops

New Haven, CT

May 2023 – May 2024

Project Liquid: Chief Engineer (2023-24), Co-President (2022-23), Propulsion Lead (2021-22) *Sept 2021 – June 2025*

- Led the development of a regeneratively-cooled liquid rocket using nitrous oxide and ethane as propellants
- Managed the design, manufacturing, and testing of a custom spark ignitor
- Designed a coaxial shear injector using a stacked manifold design as Propulsion Lead

CubeSat Team – Attitude Determination and Control System (ADCS) Lead (2022-23)

Sept 2021 – June 2023

- Led the ADCS Subteam to prepare CubeSat for launch as part of NASA's CubeSat Launch Initiative
- Designed and implemented an algorithm to control CubeSat's active stabilization efforts using magnetorquers

TECHNICAL SKILLS

Programming Languages: C, C++, D, Java, Python, MATLAB, Julia, Rust

CAD: Solidworks, Autodesk Inventor, Fusion360, OnShape

Manufacturing: Lathe, Mill, CNC, 3D-Printer, Hand Tools

Software: Project Management (Confluence/Jira), Collaboration (GitLab, GitHub, Solidworks PDM), Microsoft Office